

Hojun Yang

M.S. Researcher — Mining Robotics & Field Autonomy

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RESEARCH INTERESTS

Field robotics and automation for mining and hazardous environments: GPS-denied underground exploration and navigation; mobile manipulation and robot learning for mining-equipment automation; UAV-based spatial AI and photogrammetry; and planetary surface robotics and in-situ resource utilization (ISRU).

EDUCATION

M.S. in Energy Resources and Geosystems Engineering

Mar 2025 – Expected Feb 2027

Sejong University, Seoul, South Korea · SMART-X Lab · Advisor: Prof. Yosoon Choi

Transferred with advisor from Pukyong National University (Sept 2025).

GPA: 4.30 / 4.5

B.S. in Energy Resources Engineering

Mar 2020 – Feb 2025

Pukyong National University, Busan, South Korea · Advisor: Prof. Yosoon Choi

GPA: 4.34 / 4.5 (Class rank: 1 of 13)

JOURNAL PUBLICATIONS

Peer-reviewed; name in bold. 4 of 6 as first author.

Yang, H., & Choi, Y. (2026). Generation of vehicle-free digital elevation models in open-pit mines using UAV image inpainting. *Measurement*, 260, 119902. [SCIE]

Yang, H., Kim, S., & Choi, Y. (2025). Development of a drone- and AI-based framework for exterior inspection and damage classification of tower cranes. *Tunnel and Underground Space*. [first author]

Yang, H., Kim, S., & Choi, Y. (2025). Review of UAV-AI-based automated visual inspection technologies and their applicability to high-risk construction machinery. *Journal of the Korean Society of Mineral and Energy Resources Engineers*. [first author, review]

Jang, M., **Yang, H.**, & Choi, Y. (2025). Development of augmented reality-based educational content for mine workers. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Choi, S., **Yang, H.**, Kim, Y., & Choi, Y. (2025). Router-drop-based robot teleoperation system for unconnected underground mines. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Yang, H., & Choi, Y. (2025). Analysis of wheel slip and path errors based on steering mechanisms of a small autonomous robot in underground rough environments. *Tunnel and Underground Space*. [first author]

CONFERENCE PRESENTATIONS

Yang, H., & Choi, Y. UAV image inpainting for vehicle-free DEM generation in open-pit mines. *SME Annual Conference & Expo (MINEXCHANGE 2026)*, Salt Lake City, UT, USA, Feb 2026. (Poster)

Yang, H., et al. Scaled collaborative-manipulator platform for robot-learning-based autonomous mining equipment. *KSMER Spring Conference*, Gyeongju, Korea, 2026. (Poster)

Yang, H., & Choi, Y. UAV image inpainting for open-pit mine DEM reconstruction. *KSMER Fall Conference*, Busan, Korea, 2025. (Oral) — **Best Paper Presentation Award**

Yang, H., & Choi, Y. Removing heavy equipment from open-pit mine DEMs via deep inpainting. *KSMER Spring Conference*, Jeju, Korea, 2025. (Oral)

Yang, H., & Choi, Y. Wheel slip and path error analysis of steering mechanisms in underground rough terrain. *KSMER Fall Joint Conference*, High1 Resort, Gangwon, Korea, 2024. (Oral)

AWARDS & HONORS

Best Paper Presentation Award, KSMER Fall Conference	2025
Gold Prize, Mine-Tech Festa (KOMIR) — GeoInpainter	2025
President's Award, Korean Society of Mineral and Energy Resources Engineers	2025
Commendation, Korean Association of Geographic Information Studies	2025
Grand Prize, Mine-Tech Festa (KOMIR) — MeshRover Connect	2024
Excellence Award, Busan Future Scientist Award (Busan Sci. & Tech. Council)	2024
Excellence Award, Geoscience Data Utilization Competition (KIGAM)	2023
Excellence Award, Mine-Tech Festa (KOMIR) — AR safety content	2023

RESEARCH EXPERIENCE

Graduate Researcher — SMART-X Lab, Sejong University (prev. Pukyong National University)

- **GeoInpainter — Vehicle-free DEM generation:** built an automated pipeline (YOLOv8 detection → SAM masking → LaMa/MAT/DeepFillv2 inpainting → SfM) that removes heavy equipment from UAV imagery before 3D reconstruction, cutting elevation error from MAE 1.276 m to 0.111 m for a loader. First-author SCIE paper (Measurement, 2026).
- **Mobile manipulation via vision-language-action models:** trained a UGV + 6-DoF collaborative arm for autonomous pick-and-place; forward-kinematics-based dataset auditing raised success to 90% (95-episode dataset). Current M.S. thesis direction.
- **MeshRover Connect — Underground teleoperation:** co-developed a quadruped that autonomously drops mesh-network relay nodes to extend teleoperation in communication-denied mines; demonstrated in an active limestone mine. Grand Prize, Mine-Tech Festa 2024.
- **Steering & wheel-slip analysis:** compared Ackermann, differential, and tracked steering on rough mine terrain using laser-grid ground truth plus simulation (best path RMSE 0.2311 m). First-author paper.
- **Drone + AI inspection of high-risk machinery:** three-stage parallel YOLO architecture (ROI → part segmentation → defect detection) for UAV-based tower-crane inspection (defect mAP 0.937); government (MOLIT)-funded.

TECHNICAL SKILLS

Robotics & autonomy: ROS (Noetic); LiDAR SLAM (FAST-LIO/FAST-LIO2, LeGO-LOAM); elevation mapping & traversability analysis; MoveIt; behavior trees; Gazebo simulation

Perception & deep learning: YOLOv8/v11; SAM; image inpainting (LaMa, MAT, DeepFillv2); OpenCV; PyTorch

Robot learning: imitation learning; ACT; SmolVLA / LeRobot

Photogrammetry & GIS: UAV mapping; Structure-from-Motion (Pix4D, Metashape); DEM & point-cloud analysis

Hardware & platforms: AgileX Scout Mini / LIMO; Unitree Go1; MyCobot Pro 600; Velodyne VLP-16; Intel RealSense D435i/D455; Xsens IMU; NVIDIA Jetson AGX Orin

Programming: Python (primary); Linux/shell

LANGUAGES

Korean (native); English (research & professional working proficiency)

REFERENCES

Available upon request.